Permutations and Combinations

ANSWER KEY

Section 1: Fundamental Counting and Linear Permutations

1 In how many ways can 5 different books be arranged on a shelf?

a. 120 ✓

b. 24

c. 60

d. 720

$5! = 5 \times 4 \times 3 \times 2 \times 1 = 120.$

2 Evaluate the number of permutations of 7 distinct items taken 3 at a time, ${}^{7}P_{3}$.

a. 210 ✓

b. 35

c. 42

d. 840

${}^{7}P_{3} = \frac{7!}{(7-3)!} = 7 \times 6 \times 5 = 210.$

3 How many 4-digit even numbers can be formed using digits $\{1, 2, 3, 5, 7, 8\}$ without repetition?

a. 120 ✓

b. 60

c. 240

d. 180

Last digit must be 2 or 8 (2 choices). First three digits from remaining 5 digits: ${}^{5}P_{3} = 60$. Total $= 60 \times 2 = 120.$

Section 2: Permutations with Repetitions and Identical Items

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4 Find the number of distinct permutations of all the letters in the word PARALLEL.

a. 3360 ✓

b. 6720

c. 1680

d. 40320

8 letters total: 3 L's, 2 A's, 1 P, 1 R, 1 E. Total $= \frac{8!}{3!\,2!} = \frac{40320}{12} = 3360$.

5 How many 6-letter arrangements of PEPPER begin with the letter P?

a. 30 ✓

b. 60

c. 20

d. 120

Fix one P first. Remaining 5 letters: 2 P's, 2 E's, 1 R. Arrangements $= \frac{5!}{2!\,2!} = \frac{120}{4} = 30$.

6 How many 5-digit codes can be formed from digits $\{3, 3, 3, 7, 7\}$?

a. 10 ✓

b. 20

c. 5

d. 120

$\frac{5!}{3!\,2!} = \frac{120}{6 \times 2} = 10$.

Section 3: Combinations and Committee Selections

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7 A committee of 4 is to be chosen from 9 people. How many distinct committees can be formed?

a. 126 ✓

b. 3024

c. 84

d. 252

${}^{9}C_{4} = \frac{9 \times 8 \times 7 \times 6}{4 \times 3 \times 2 \times 1} = 126.$

8 A team of 5 is selected from 6 boys and 4 girls. How many teams contain exactly 3 boys and 2 girls?

a. 120 ✓

b. 60

c. 210

d. 100

Choose boys: $\binom{6}{3} = 20$. Choose girls: $\binom{4}{2} = 6$. Total $= 20 \times 6 = 120.$

9 From 10 points on a plane where no three are collinear, how many distinct triangles can be drawn?

a. 120 ✓

b. 720

c. 90

d. 45

${}^{10}C_{3} = \frac{10 \times 9 \times 8}{3 \times 2 \times 1} = 120.$

Section 4: Restricted Arrangements and Separations

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10 In how many ways can 4 boys and 3 girls sit in a row if all 3 girls must sit together?

a. 720 ✓

b. 5040

c. 144

d. 288

Treat 3 girls as 1 block. Total units: 4 boys + 1 block = 5 units ($5! = 120$). Girls within block: $3! = 6$. Total $= 120 \times 6 = 720$.

11 In how many ways can 5 people sit around a circular dinner table?

a. 24 ✓

b. 120

c. 60

d. 12

Circular permutations: $(n-1)! = (5-1)! = 4! = 24$.

12 A committee of 4 is selected from 5 men and 5 women. How many committees contain at least one woman?

a. 205 ✓

b. 210

c. 195

d. 150

Total selections $= \binom{10}{4} = 210$. All-men selections $= \binom{5}{4} = 5$. At least one woman $= 210 - 5 = 205$.

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